

Nityanand Mathur

 [Github](#)  [LinkedIn](#)  nityanandmathur@gmail.com  [7247412358](tel:7247412358)  [Google Scholar](#)

EDUCATION

Indian Institute Of Information Technology Guwahati

B. Tech in Computer Science & Engineering

GPA: 8.16/10

PUBLICATIONS

CLIPDrawX: Primitive-based Explanations for Text Guided Sketch Synthesis

Nityanand Mathur, Shyam Marjit, Abhra Chaudhuri, Anjan Dutta

 [Project](#)

 [Paper](#)

DiffuseKronA: Param. Efficient Finetuning Method for Personalized Diff'sn Models

Shyam Marjit, Harshit Singh, Nityanand Mathur, Sayak Paul, Chia-Mu Yu, Pin-Yu Chen

 [Project](#)

 [WACV](#)

SVGCraft: Beyond Single Object Text-to-SVG Syn. with Compr've Canvas Layout

Ayan Banerjee, Nityanand Mathur, Josep Lladós, Umapada Pal, Anjan Dutta

 [Project](#)

 [Paper](#)

WORK EXPERIENCE

Smallest AI | *Data Scientist*

October 2024 - Present

- Developed a real-time, flow-based instant voice cloning model. Built professional voice cloning using LoRA.
- Optimized a TTS model, reducing latency from 300–400ms to 150–250ms, improving real-time performance.
- Implemented an end-to-end text normalization module to handle emails, websites, dates, numbers, etc.

Bosch Research | *Machine Learning Intern*

Jan 2024 - August 2024

- Created novel algorithms for synthetic dataset generation by object augmentation using latent diffusion models.
- Created parameter-efficient label preserving source-free domain adaptation and localised editing pipelines.

University of Surrey | *Research Intern - Dr. Anjan Dutta* |  [Project Page](#) |  [Paper](#) Jan, 2023 – December, 2023

- Worked on introducing explainability to CLIP-based models using simple primitives, with an LDM-powered initialization for faster convergence. Introduced *Primitive-level Dropout* for noiseless sketch synthesis.

IBM | *Research Intern - Dr. Pin-Yu Chen* |  [Project Page](#) |  [Paper](#)

June, 2023 - December, 2023

- Worked on adding parameter-efficient *Kronecker Product* based adapters to personalized T2I models that are ~35% more efficient than SOTA, while generating images with high fidelity and text-alignment.

Osaka University | *Data Science Research Intern — Dr. Manas Kala* |  

July, 2023 - Sept, 2023

- Applied counterfactual machine learning to thermal comfort dataset to simulate the comfort of students in winter conditions to find the impact of clothing, age, grade and gender. — *Work undergoing at univ.*

CogXR Labs | *Computer Vision, MLOps Intern*

Nov, 2022 – March, 2023

- Implemented large-scale image classification algorithms on healthcare datasets with high accuracy.
- Created end-to-end production pipelines using docker containers, DVC, W&B and Pytest

IIIT Guwahati | *Research Intern — Dr. Ferdous Ahmed Barbhuiya* | 

May, 2022 - August, 2022

- Implemented VisualBERT-based multimodal hateful meme classification on social media.
- Integrated CLIP-based embeddings to improve accuracy from 75% to 81%.

IIIT Guwahati | *Research Intern — Dr. Radhika Sukapuram* | 

May, 2022 – July, 2022

- Researched service caching algorithms in edge cloud to maximize the cache hit rate at edge devices.

PROJECTS

DINO Explorer  | *Python, Computer Vision*

June 2024

- Created a powerful tool designed to explore and visualize DINOv2 embeddings. Given a list of folders containing images, DINO Explorer extracts their DINO embeddings and creates an interactive visualization using Voxel51.

CVP  | *Python, Pytorch, Segmentation, GANs, Neural Style Transfer*

January 2023

- Implemented U-net based image segmentation for humans & aerial image segmentation to detect roads in a map
- Implemented object localization for food items and neural style transfer using Efficient-net.

CheXpert  | *Python, Tensorflow, Computer Vision, Docker, DVC, W&B, Git*

November 2022

- Implemented a deep learning multi-label classifier to process an 11 GB medical image dataset, classifying X-ray images into 13 disease categories. Later, merged with the Classify-Covid project to add Covid.

Classify-Covid   | *Python, Tensorflow, Computer Vision, Docker, DVC, W&B, Git*

August 2022

- Developed a binary classification model using DenseNet architecture to analyze a substantial X-ray dataset for COVID-19 detection, with 82% accuracy. Containerized using docker dev containers.

SKILLS

Languages: Python, SQL, Bash, C, Java, L^AT_EX

Frameworks/Libraries: PyTorch, TensorFlow, Keras, Pandas, NumPy, Scikit Learn, and OpenCV

Tools: Docker, Git/GitHub, Unix Shell, PyTest, Weights and Biases, DVC, Hydra.cc, Hugging Face, AWS, Gradio